

FOUNDATIONS OF AI (FAI)

PROJECT REPORT

Border Threat Detection Expert System

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SECTION 1 — INTRODUCTION AND DOMAIN DEFINITION

- Artificial Intelligence (AI) is one of the most important technologies used in modern systems for decision-making, automation, monitoring, and problem-solving. One of the major branches of Artificial Intelligence is the Expert System. An expert system is a knowledge-based AI system that uses stored facts, logical rules, and reasoning techniques to make decisions similar to a human expert. These systems are designed to solve specific problems by analyzing information and generating conclusions based on predefined knowledge. Expert systems are widely used in various fields such as healthcare, banking, education, transportation, cybersecurity, and defense. They help organizations improve efficiency, reduce human workload, and support accurate decision-making. Unlike traditional software systems, expert systems focus on reasoning and logical inference rather than simple calculations. They use a knowledge base and inference engine to process information and generate recommendations.
- The selected domain for this project is Border Security and Threat Detection in defense operations. Border security is an extremely important aspect of national defense because unauthorized border activities can create serious security risks for a country. Borders are continuously monitored to detect illegal entry, suspicious movement, smuggling, infiltration attempts, and other dangerous activities that may threaten national safety and stability. In many situations, monitoring large border areas manually becomes difficult for security personnel because border regions may cover long distances, isolated terrains, mountains, deserts, forests, and remote areas. Continuous surveillance requires significant manpower, time, and resources. Human monitoring can also be affected by fatigue, delayed response, and limited visibility during night operations or harsh weather conditions.
- To improve border monitoring and threat detection, intelligent systems based on Artificial Intelligence can be used to assist defense personnel. The Border Threat Detection Expert System is designed as a theoretical AI-based expert system that helps identify suspicious activities and possible threats using logical reasoning and predefined rules. The system analyzes observed conditions such as motion detection, thermal signal detection, unauthorized border crossing, repeated movement patterns, and night-time activity to determine the threat level of a situation. The main objective of this expert system is to support defense and border security personnel in making faster and more accurate decisions. Instead of relying only on manual observation, the system uses stored knowledge and logical IF-THEN rules to automatically evaluate situations and generate recommendations. The expert system can classify situations as low threat, medium threat, or high threat depending on the available evidence and detected activities.
- For example, if the system detects movement during night hours along with unauthorized entry signals, it can identify the situation as suspicious and generate a high-threat alert. Similarly, if thermal signals are detected repeatedly in restricted areas, the system may recommend deploying surveillance drones or sending immediate security alerts for further investigation.
- The expert system works using symbolic AI concepts and knowledge representation techniques. Information is represented in the form of facts and logical conditions, while the decision-making process is controlled through an inference mechanism. By applying logical rules step-by-step, the system derives conclusions and provides recommendations similar to those given by experienced security experts. This project demonstrates how expert systems can be applied in defense-related environments to improve monitoring efficiency, assist in threat identification, and support strategic decision-making. Although the system presented in this report is theoretical and does not involve programming implementation, it effectively represents the core concepts of knowledge representation, rule-based reasoning, and logical inference used in Artificial Intelligence systems. The final output of the Border Threat Detection Expert System includes generating security alerts, identifying threat levels, recommending drone surveillance deployment, and assisting defense authorities in responding quickly to suspicious activities near border regions. The system aims to reduce response time, improve monitoring accuracy, and enhance border security operations through intelligent reasoning techniques.



SECTION 2 — KNOWLEDGE REPRESENTATION

Content

- The expert system represents border security knowledge using logical facts, properties, and conditions.

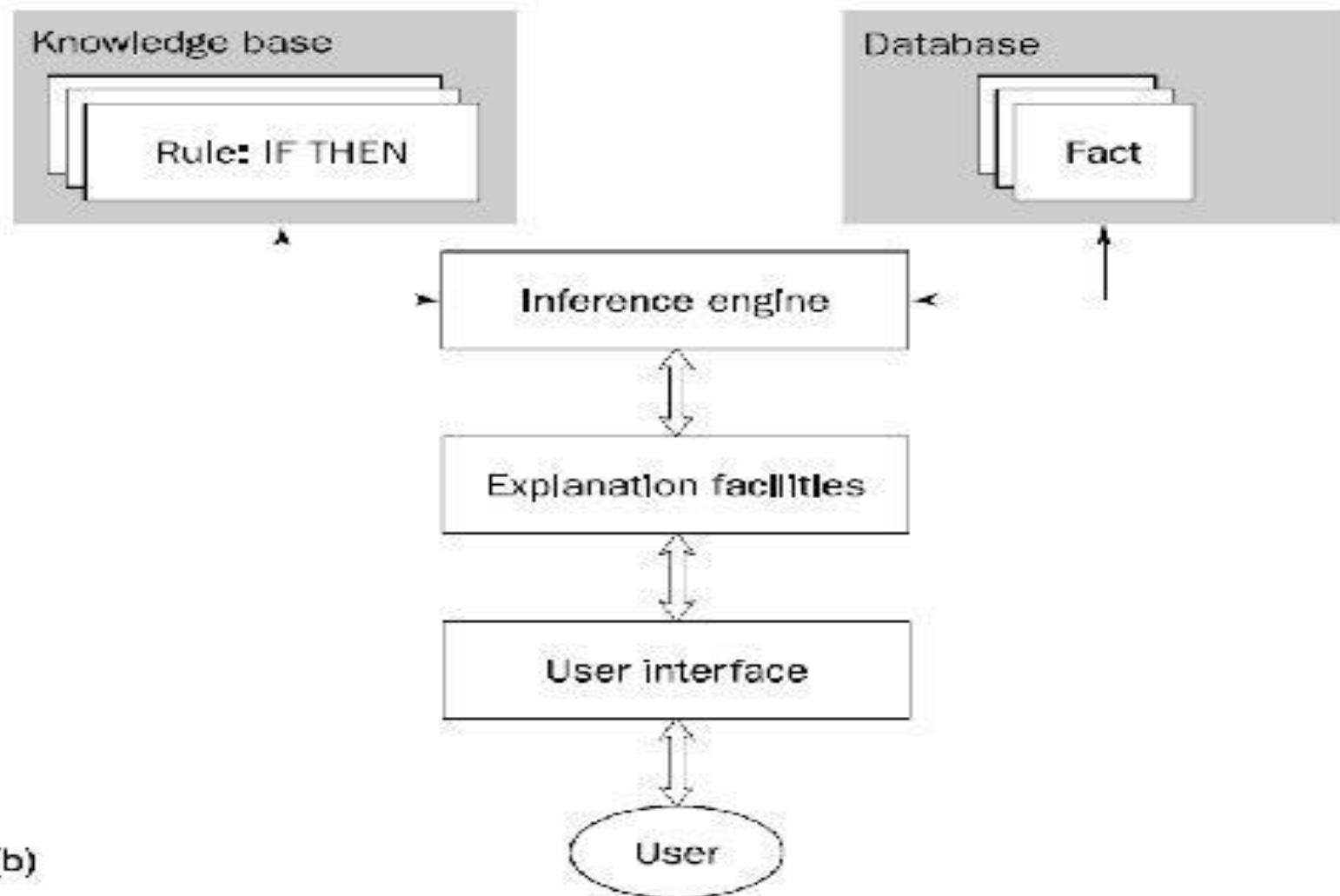
Key Concepts:

- Motion Detection
- Thermal Signal Detection
- Unauthorized Entry
- Night-Time Activity
- Repeated Border Crossing
- Drone Surveillance
- Threat Level

Formal Representation:

Motion Detected(Person)
Thermal Signal Detected(Person)
Unauthorized Entry(Person)
Night Activity(Person)
Repeated Movement(Person)
Suspicious Activity(Person)
High Threat(Person)
Medium Threat(Person)
Deploy Drone(Person)
Send Alert(Person)

Basic Structure of Rule Based Expert System



SECTION 3 — THE KNOWLEDGE BASE: RULES

RULE 1:

IF Motion Detected(Person) AND Night Activity(Person)

THEN Suspicious Activity(Person)

RULE 2:

IF Thermal Signal Detected(Person)

THEN Medium Threat(Person)

RULE 3:

IF Unauthorized Entry(Person)

THEN High Threat(Person)

RULE 4:

IF Repeated Movement(Person)

THEN Suspicious Activity(Person)

RULE 5:

IF Suspicious Activity(Person) AND Thermal Signal Detected(Person)

THEN High Threat(Person)

RULE 6:

IF High Threat(Person)
THEN Send Alert(Person)

RULE 7:

IF High Threat(Person)
THEN Deploy Drone(Person)

RULE 8:

IF Medium Threat(Person) AND Night Activity(Person)
THEN Send Alert(Person)

RULE 9:

IF Unauthorized Entry(Person) AND Thermal Signal Detected(Person)
THEN Deploy Drone(Person)

RULE 10:

IF Suspicious Activity(Person)
THEN Medium Threat(Person)

SECTION 4 — INFERENCE STRATEGY

This expert system uses Forward Chaining as its inference strategy.

Forward Chaining is a data-driven reasoning process that starts with known facts and applies logical rules to derive new conclusions. This strategy is suitable for border threat detection because the system continuously receives observed data such as movement detection, thermal signals, and unauthorized entry reports.

Using Forward Chaining allows the system to automatically generate alerts and recommendations as soon as new information becomes available.

SECTION 5 — SAMPLE CONSULTATION: STEP-BY-STEP TRACE

- **STEP 1 — INITIAL FACTS**

- Known Facts:

Motion Detected(Person1)

Night Activity(Person1)

Thermal Signal Detected(Person1)

Unauthorized Entry(Person1)

- **STEP 2 — TRACE PROCESS**

- Step 1:

The system identifies Motion Detected(Person1) and Night Activity(Person1).

RULE 1 fires:

IF Motion Detected(Person) AND Night Activity(Person)

THEN Suspicious Activity(Person)

New Fact Derived:

Suspicious Activity(Person1)

Step 2:

The system detects Thermal Signal Detected(Person1).

RULE 2 fires:

IF Thermal Signal Detected(Person)

THEN Medium Threat(Person)

New Fact Derived:

Medium Threat(Person1)

Step 3:

The system detects Unauthorized Entry(Person1).

RULE 3 fires:

IF Unauthorized Entry(Person)

THEN High Threat(Person)

New Fact Derived:

High Threat(Person1)

Step 4:

The system now has:

Suspicious Activity(Person1)

and

Thermal Signal Detected(Person1)

RULE 5 fires:

IF Suspicious Activity(Person) AND Thermal Signal Detected(Person)

THEN High Threat(Person)

New Fact Confirmed:

High Threat(Person1)

Step 5:

The system identifies High Threat(Person1).

RULE 6 fires:

IF High Threat(Person)

THEN Send Alert(Person)

New Fact Derived:

Send Alert(Person1)

Step 6:

RULE 7 fires:

IF High Threat(Person)

THEN Deploy Drone(Person)

New Fact Derived:

Deploy Drone(Person1)

Final Conclusion:

The system concludes that Person1 represents a high border threat. Therefore, an alert is generated and drone surveillance is deployed for further monitoring and security action.



SECTION 6 — CONCLUSION AND LIMITATIONS

The Border Threat Detection Expert System demonstrates how symbolic AI and rule-based reasoning can be applied in defense and border security operations. The system uses logical IF-THEN rules and Forward Chaining inference to identify suspicious activities and generate security recommendations.

The expert system can help improve monitoring efficiency and assist defense personnel in decision-making.

However, the system has certain limitations. It cannot handle uncertain or incomplete information effectively. The system also depends completely on predefined rules and may produce incorrect results if the input facts are inaccurate. Additionally, conflicting rules may create ambiguity in decision-making.

